

Differential Equations

16

SYLLABUS

5.18

5.19

HL ONLY

A cup of coffee cools fastest when it is hottest. Pull it from the machine at 80 degrees and it drops quickly at first; near room temperature it barely changes. The *rate* at which it cools depends on how far above the room its temperature currently is.

That sentence is a differential equation. Written in symbols, the rate of cooling $\frac{d\theta}{dt}$ is proportional to the gap $\theta - 20$ between the coffee and the 20-degree room:

$$\frac{d\theta}{dt} = -k(\theta - 20).$$

Notice what kind of equation this is. The unknown is not a number. It is a *function*, $\theta(t)$, and the equation links that function to its own derivative. Most laws of nature have this form, because what we can measure directly is usually a rate: how fast something grows, cools, decays or moves.

A **differential equation** relates a function to its derivatives. To **solve** one is to find the function itself, recovering $\theta(t)$ from a statement about $\frac{d\theta}{dt}$.

That is the point of solving one: it turns a rule about change into a prediction. A law of nature tells you what is happening at this instant; solving its differential equation tells you the whole future.

By the end of this chapter you will be able to solve such an equation exactly by three methods: separating the variables, substituting to make them separate, and using an integrating factor. You will also write a function as an infinite polynomial, called a Maclaurin series. That gives a fourth route to a solution, and a way to approximate a function. When no exact solution exists, you will solve the equation approximately by Euler's method. The coffee equation is solved in the first section.

A calculator is assumed throughout this chapter. The **EXACT** badge marks the questions whose answer must be left exact.

16.1 Separable Variables

Differential equations appear wherever a law describes a rate of change, which covers most of science. A few examples show the range.

- **Cooling.** $\frac{d\theta}{dt} = -k(\theta - \theta_{\text{room}})$ — a hot drink, an engine, or a body found at a crime scene.
- **Radioactive decay.** $\frac{dN}{dt} = -\lambda N$ — the basis of carbon dating.
- **Population growth.** $\frac{dn}{dt} = kn(M - n)$ — growth that slows as the population fills the space available.
- **Medicine.** $\frac{dC}{dt} = -kC$ — a drug leaving the bloodstream, which sets how often a dose must be repeated.
- **Falling through air.** $\frac{dv}{dt} = g - kv^2$ — why a skydiver reaches a steady terminal speed.
- **Epidemics.** Three linked differential equations predict how an infection spreads through a population.

Each one says the same thing in different words: the rate is known, and the quantity is wanted.

Two words describe the equations in this chapter, and both are settled by looking at the equation.

- The **order** is the highest derivative that appears. Every equation here contains $\frac{dy}{dx}$ and no higher derivative, so all of them are **first order**.
- An equation is **linear** when y and its derivatives appear only to the first power, and are never multiplied by one another. So $\frac{dy}{dx} + xy = 1$ is linear, and $\frac{dy}{dx} = 2xy^2$ is not.

The three exact methods in this chapter are chosen by which form an equation takes, so these two points are checked first.

16.1.1 Separating the variables

The simplest case can be rearranged so that each variable sits on its own side. A first-order equation is **separable** if it can be written as a function of y times a function of x :

$$\frac{dy}{dx} = g(x) h(y).$$

Then you separate and integrate both sides. Treating $\frac{dy}{dx}$ as a ratio is informal but valid here: gather all the y terms with dy and all the x terms with dx .

KEY

$$\frac{dy}{dx} = g(x)h(y) \implies \int \frac{1}{h(y)} dy = \int g(x) dx$$

One arbitrary constant appears, giving the **general solution**, a family of curves. An **initial condition** fixes the constant and selects the **particular solution**.

EXAMPLE 16.1

Solve $\frac{dy}{dx} = 2xy^2$ given $y(0) = 1$.

Separate the variables and integrate:

$$\int \frac{1}{y^2} dy = \int 2x dx \implies -\frac{1}{y} = x^2 + C.$$

Apply $y(0) = 1$: $-1 = 0 + C \implies C = -1$. So $-\frac{1}{y} = x^2 - 1$, giving

$$y = \frac{1}{1 - x^2}.$$

Always apply the initial condition as early as possible; carrying C further than necessary invites errors.

EXAMPLE 16.2

A cup of coffee obeys $\frac{d\theta}{dt} = -k(\theta - 20)$ with $\theta(0) = 80$. Solve for $\theta(t)$.

Separate and integrate:

$$\int \frac{1}{\theta - 20} d\theta = \int -k dt \implies \ln |\theta - 20| = -kt + C.$$

Exponentiating, $\theta - 20 = Ae^{-kt}$. The condition $\theta(0) = 80$ gives $A = 60$, so

$$\theta(t) = 20 + 60e^{-kt}.$$

The temperature decays exponentially toward the room's 20 degrees, never quite reaching it: exactly the behaviour we observed.

Separation also handles the most famous population model in ecology. In the 1960s, once the cattle disease rinderpest was eliminated around the Serengeti, the wildebeest herd began to grow, and its growth obeyed the **logistic equation**: growth proportional both to the herd itself and to the room left below the plains' carrying capacity.

EXAMPLE 16.3

The Serengeti wildebeest population n (in millions) satisfies $\frac{dn}{dt} = kn(1.4 - n)$, with $n(0) = 0.25$ in 1963. Solve for $n(t)$.

Separate, and split the left side with the partial fractions of Ch. 6:

$$\int \frac{dn}{n(1.4 - n)} = \int k dt, \quad \frac{1}{n(1.4 - n)} = \frac{1}{1.4} \left(\frac{1}{n} + \frac{1}{1.4 - n} \right).$$

Integrating,

$$\frac{1}{1.4} \ln \left| \frac{n}{1.4 - n} \right| = kt + C \Rightarrow \frac{n}{1.4 - n} = Be^{1.4kt}.$$

The initial condition gives $B = \frac{0.25}{1.15} \approx 0.217$, and solving for n :

$$n(t) = \frac{1.4}{1 + 4.6 e^{-1.4kt}}.$$

This is the logistic S-curve: growth that is nearly exponential at first, then levels off at the carrying capacity of 1.4 million. The bracket $(1.4 - n)$ is what slows it, since it shrinks towards zero as n approaches 1.4. The real herd behaved this way, climbing to about 1.3 million by 1977 and staying near that level since.

YOUR TURN 16.1 Separable Variables

1. Find the general solution of each equation.

(a) $\frac{dy}{dx} = \frac{x}{y}$

(b) $\frac{dy}{dx} = y$

(c) $\frac{dy}{dx} = \frac{1 + y}{1 + x}$

(d) $\frac{dy}{dx} = e^{x-y}$

2. Find the particular solution of each equation.

(a) $\frac{dy}{dx} = xy$ with $y(0) = 2$.

(b) $\frac{dy}{dx} = 3x^2y$ with $y(0) = 1$.

(c) $\frac{dy}{dx} = y \tan x$ with $y(0) = 4$, for $0 \leq x < \frac{\pi}{2}$.

(d) Explain why a general solution contains one arbitrary constant, and what the initial condition does to it.

16.2 Homogeneous Differential Equations

Some equations cannot be separated as they stand, but a substitution makes them separable.

A first-order equation is **homogeneous** if the right-hand side depends only on the ratio $\frac{y}{x}$, and not on x and y separately.

Equations of this kind appear when a problem has no fixed size, only directions and proportions.

- **Pursuit curves.** A ferry always aims at the dock while a current pushes it sideways. At each moment only the *direction* from the ferry to the dock matters, and that direction is set by $\frac{y}{x}$ alone. You will solve the ferry in the challenge problems.
- **Reflector shapes.** The curve of a car headlight or a satellite dish is fixed by requiring every ray from the focus to leave parallel. That condition involves only angles, so the equation depends on the ratio.
- **Scale-free rules.** If doubling both quantities leaves a rule unchanged, the rule can only involve their ratio, and the equation is homogeneous.

The substitution $y = vx$ turns such an equation into a separable one. Since $y = vx$, the product rule gives $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

KEY For $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$, substitute $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

EXAMPLE 16.4

Solve $\frac{dy}{dx} = \frac{x+y}{x}$.

Rewrite the right side in terms of $\frac{y}{x}$: $\frac{x+y}{x} = 1 + \frac{y}{x}$. Substitute $y = vx$, $\frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$v + x \frac{dv}{dx} = 1 + v \implies x \frac{dv}{dx} = 1.$$

This is separable:

$$\int dv = \int \frac{1}{x} dx \implies v = \ln|x| + C.$$

Finally replace $v = \frac{y}{x}$:

$$y = x(\ln|x| + C).$$

Here the single v on each side cancelled, but nothing depends on that. Whatever f is, the substitution turns the equation into $x \frac{dv}{dx} = f(v) - v$, with v on one side and x on the other. It is separable however $f(v) - v$ turns out. That is what makes the substitution work.

EXAMPLE 16.5

Solve $\frac{dy}{dx} = \frac{2x^2 + y^2}{xy}$ for $x > 0$.

Divide the top and bottom by x^2 to show that the right side depends only on the ratio:

$$\frac{2x^2 + y^2}{xy} = \frac{2 + (y/x)^2}{y/x}.$$

Substitute $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$:

$$v + x \frac{dv}{dx} = \frac{2 + v^2}{v} \implies x \frac{dv}{dx} = \frac{2 + v^2}{v} - v = \frac{2}{v}.$$

Again the v terms cancel, leaving a separable equation:

$$\int v dv = \int \frac{2}{x} dx \implies \frac{v^2}{2} = 2 \ln x + C.$$

Replace $v = \frac{y}{x}$ and clear the fraction:

$$y^2 = 2x^2(2 \ln x + C).$$

Checking that the right side really depends only on $\frac{y}{x}$ is the first step every time. Dividing top and bottom by the highest power of x is the quickest way to see it.

YOUR TURN 16.2 Homogeneous Differential Equations

3. Use the substitution $y = vx$ to solve each equation.

(a) $\frac{dy}{dx} = \frac{2x + y}{x}$

(b) $\frac{dy}{dx} = \frac{y - x}{x}$

(c) $\frac{dy}{dx} = \frac{x + 2y}{x}$

(d) $\frac{dy}{dx} = \frac{y}{x}$ for $x > 0$.

4.

(a) Show that $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ is homogeneous, by writing the right-hand side in terms of $\frac{y}{x}$.

(b) Hence find its general solution.

16.3 The Integrating Factor

A **linear** first-order differential equation has the form

$$\frac{dy}{dx} + P(x)y = Q(x).$$

It is usually not separable. The method is to multiply through by a carefully chosen function, the **integrating factor**, which turns the left side into the derivative of a single product.

KEY · IB For $\frac{dy}{dx} + P(x)y = Q(x)$, the integrating factor is

$$I = e^{\int P(x) dx}.$$

After multiplying by I , the left side becomes exactly $\frac{d}{dx}(Iy)$, by the product rule run backwards. You then integrate both sides directly.

EXAMPLE 16.6

Solve $\frac{dy}{dx} + \frac{1}{x}y = x^2$ for $x > 0$.

Here $P(x) = \frac{1}{x}$, so the integrating factor is

$$I = e^{\int \frac{1}{x} dx} = e^{\ln x} = x.$$

Multiply through by x :

$$x \frac{dy}{dx} + y = x^3 \implies \frac{d}{dx}(xy) = x^3.$$

The left side has collapsed to a single derivative. Integrate both sides:

$$xy = \frac{x^4}{4} + C \implies y = \frac{x^3}{4} + \frac{C}{x}.$$

The method exists to create that $\frac{d}{dx}(xy)$ on the left; everything after is ordinary integration.

EXAMPLE 16.7

Solve $\frac{dy}{dx} + y = e^x$.

$P(x) = 1$, so $I = e^{\int 1 dx} = e^x$. Multiply through:

$$\frac{d}{dx}(e^x y) = e^x \cdot e^x = e^{2x}.$$

Integrate: $e^x y = \frac{1}{2}e^{2x} + C$, hence $y = \frac{1}{2}e^x + Ce^{-x}$.

16.3.1 Where linear equations come from

A linear equation appears whenever a quantity is added and removed at the same time, and either rate may change.

- **Mixing tanks.** Liquid flows into a tank and drains out of it, and the dissolved substance obeys a linear equation. When the volume itself changes, the equation is linear but not separable, so the integrating factor is the method that applies.
- **Drug infusion.** A drug enters the blood at a steady rate through a drip, while the body clears it at a rate proportional to the amount already present.
- **Electrical circuits.** In a circuit with a resistor and a capacitor, the charge satisfies a linear equation driven by the applied voltage.
- **Savings with deposits.** An account earns interest continuously while money is paid in at a steady rate.

The worked example below is a mixing problem of the first kind.

EXAMPLE 16.8

A 100-litre vat in a brewery holds 5 kg of dissolved sugar. Syrup carrying 0.2 kg of sugar per litre flows in at 1 litre per minute, the vat is kept well stirred, and the mixture drains at 2 litres per minute. Find the sugar $S(t)$ in the vat at time t minutes.

The volume shrinks: $V(t) = 100 - t$. Sugar arrives at $0.2 \times 1 = 0.2 \text{ kg min}^{-1}$ and leaves at concentration times outflow, $\frac{S}{100-t} \times 2$. So

$$\frac{dS}{dt} = 0.2 - \frac{2S}{100-t} \implies \frac{dS}{dt} + \frac{2}{100-t} S = 0.2.$$

The integrating factor is

$$I = e^{\int \frac{2}{100-t} dt} = e^{-2\ln(100-t)} = (100-t)^{-2}.$$

Multiplying through collapses the left side:

$$\frac{d}{dt} \left(S(100-t)^{-2} \right) = 0.2(100-t)^{-2} \implies S(100-t)^{-2} = \frac{0.2}{100-t} + C.$$

So $S = 0.2(100-t) + C(100-t)^2$, and $S(0) = 5$ gives $C = -0.0015$:

$$S(t) = 0.2(100-t) - 0.0015(100-t)^2.$$

Sense-check the endpoints: $S(0) = 5$, and $S(100) = 0$, an empty vat holds no sugar. In between, the sugar first *rises*, to about 6.7 kg near $t = 33$, before the draining takes over. That rise is not

obvious from the description, and it is the equation rather than intuition that reveals it.

16.3.2 Choosing a method

Three exact methods are now available, and a question rarely says which to use. The form of the equation decides it, and the test is quick.

What the equation looks like	Method	Example
The variables can be split, $\frac{dy}{dx} = g(x)h(y)$	separate and integrate	$\frac{dy}{dx} = 2xy^2$
The right side depends only on $\frac{y}{x}$	substitute $y = vx$	$\frac{dy}{dx} = \frac{x+y}{x}$
Linear in y : $\frac{dy}{dx} + P(x)y = Q(x)$	integrating factor $e^{\int P dx}$	$\frac{dy}{dx} + \frac{1}{x}y = x^2$
None of the above	Euler's method, numerically	$\frac{dy}{dx} = x^2 + y^2$

Try them in that order, because each test is quicker than the one after it. Note that the first two overlap: an equation can be both separable and homogeneous, in which case separating directly is the shorter route. A linear equation can overlap too. When Q/P is constant, $\frac{dy}{dx} + P(x)y = Q(x)$ rearranges to $\frac{dy}{dx} = P(x) \left(\frac{Q}{P} - y \right)$ and separates at once, which is how $\frac{dy}{dx} + y = 4$ and the cooling equation of §16.1 are solved. In general, though, Q does not divide P that way, nothing separates, and the integrating factor is what is left.

YOUR TURN 16.3 The Integrating Factor

5. Solve each linear equation.

- (a) $\frac{dy}{dx} + y = 4$
- (b) $\frac{dy}{dx} + 2y = 6$
- (c) $\frac{dy}{dx} + 2y = e^x$
- (d) $\frac{dy}{dx} - y = e^{2x}$

6.

- (a) Write down the integrating factor for $\frac{dy}{dx} + \frac{3}{x}y = x^2$, for $x > 0$.

- (b) Solve $\frac{dy}{dx} + \frac{2}{x}y = x$ for $x > 0$.
- (c) Solve $\frac{dy}{dx} + \frac{1}{x}y = x$ with $y(1) = 1$, for $x > 0$.
- (d) Explain what the integrating factor is chosen to achieve on the left-hand side.

16.4 Maclaurin Series

Many functions can be written as an infinite polynomial. This idea stands apart from the three methods above, and it has two uses: it approximates a function, and it gives a fourth route to solving a differential equation.

There is a reason a polynomial is wanted at all. A polynomial is built only from adding and multiplying, and those are the two operations a machine can actually perform. Nothing inside a calculator knows what $\sin x$ is. When you press the sine key, the calculator evaluates a polynomial instead. Every function outside that small arithmetic world must be turned into one before it can be computed, and a Maclaurin series is how that is done.

The method is to match the function at a single point. Near $x = 0$, we look for a polynomial with the same value as the function, then the same gradient, then the same concavity, then the same higher derivatives. The more derivatives agree, the longer the polynomial stays close to the curve as x moves away from 0. The result is the **Maclaurin series**.

The coefficients are not chosen; they are forced. Suppose the series exists and write it as

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

Putting $x = 0$ kills every term but the first, so $a_0 = f(0)$. Differentiating gives $f'(x) = a_1 + 2a_2x + 3a_3x^2 + \dots$, and $x = 0$ now leaves $a_1 = f'(0)$. Differentiating again gives $f''(0) = 2a_2$, and once more $f'''(0) = 6a_3 = 3!a_3$. Each round of differentiating strips one term and multiplies the next by its own index, so in general $f^{(n)}(0) = n!a_n$.

KEY · IB

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \dots$$

EXAMPLE 16.9

Find the Maclaurin series for $f(x) = e^x$.

Every derivative of e^x is e^x , and $e^0 = 1$, so $f^{(n)}(0) = 1$ for all n :

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

This series can be taken as the definition of e^x , and it computes the function to any accuracy you want. It also explains why e^x is its own derivative: differentiate the series term by term and you get the same series back.

The series below are provided in the formula booklet. Each is exact, and truncating it gives a polynomial approximation that is excellent near $x = 0$.

KEY · IB

$$\begin{aligned} e^x &= 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots & \sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots & \ln(1+x) &= x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \\ \arctan x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \end{aligned}$$

One more expansion is named by the syllabus but printed elsewhere in the booklet, so it is easy to overlook. The binomial family $(1+x)^p$ is a Maclaurin series like any other, and it is the extended binomial theorem of Ch. 2 (booklet 1.10) seen from this side.

KEY For $p \in \mathbb{Q}$ and $|x| < 1$,

$$(1+x)^p = 1 + px + \frac{p(p-1)}{2!}x^2 + \frac{p(p-1)(p-2)}{3!}x^3 + \dots$$

Unlike the series above it carries a validity condition, because it does not converge for every x .

EXAMPLE 16.10

Use a Maclaurin series to approximate $\sqrt{1.1}$ to four decimal places.

Write $\sqrt{1.1} = (1+x)^{1/2}$ with $x = 0.1$, which satisfies $|x| < 1$. With $p = \frac{1}{2}$,

$$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \dots$$

Substituting $x = 0.1$ gives

$$1 + 0.05 - 0.00125 + 0.0000625 = 1.0488125.$$

The true value is $\sqrt{1.1} = 1.0488088 \dots$, so four terms agree to four decimal places, 1.0488. Each term is smaller than the last by roughly a factor of x , which is why these series are useful: the smaller

x is, the fewer terms you need.

The figure below shows the first three odd-degree truncations of the sine series: the line $y = x$, the cubic, and the quintic. Each new term extends the interval over which the polynomial and the curve agree.

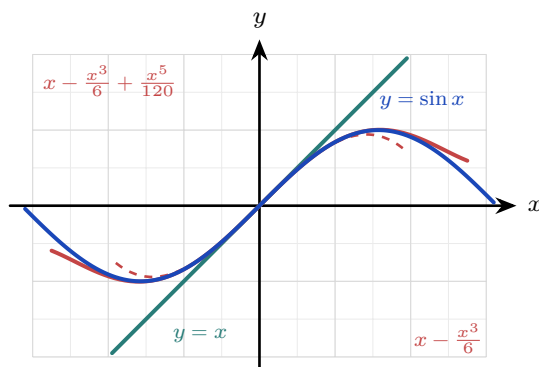


Figure 16.1 The first three odd-degree truncations of the sine series against $y = \sin x$. Each extra term widens the interval on which the polynomial and the curve agree, but every truncation eventually leaves the curve.

The same convergence can be watched numerically. Below are the successive truncations of the sine series at $x = 1$, against the true value $\sin 1 = 0.841471$.

terms up to	x	x^3	x^5	x^7
estimate	1	0.833333	0.841667	0.841468
error	0.159	0.0081	0.0002	0.0000027

Each new term cuts the error by a factor of tens, and that factor grows as more terms are taken: the ratios along this row are about 20, 40 and 70. That is the trade the series offers: a polynomial is simple enough to compute, and taking more terms buys as much accuracy as you are willing to pay for.

16.4.1 Building new series

New series are rarely built from the definition. They are made from the standard five by four operations, all of which the syllabus names.

- **Substitute.** Replacing x with $-x^2$ in the series for e^x gives

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \dots$$

- **Differentiate.** Differentiating the sine series term by term produces exactly the cosine series: the identity $\frac{d}{dx} \sin x = \cos x$ in polynomial form.
- **Integrate.** The geometric series of Ch. 2 gives $\frac{1}{1+x^2} = 1 - x^2 + x^4 - \dots$ for $|x| < 1$, and

integrating term by term yields $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots$. The booklet's arctangent series is simply the geometric series integrated.

- **Multiply.** For a product, expand each factor and collect powers up to the degree you need.

EXAMPLE 16.11

Find the Maclaurin series of $e^x \sin x$ up to the term in x^3 .

Multiply the two standard series, keeping only terms of degree ≤ 3 :

$$\left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right) \left(x - \frac{x^3}{6}\right) = x + x^2 + \left(\frac{1}{2} - \frac{1}{6}\right)x^3 + \dots = x + x^2 + \frac{x^3}{3} + \dots$$

The work is bookkeeping. Decide the target degree first, and discard anything above it before multiplying rather than after.

16.4.2 Small angles and series from differential equations

Truncating at the first term or two gives the **small-angle approximations** used throughout physics. For θ near 0, in radians,

$$\sin \theta \approx \theta, \quad \cos \theta \approx 1 - \frac{\theta^2}{2}, \quad \tan \theta \approx \theta.$$

These are not conveniences invented by physicists; they are the first one or two terms of the series, with the rest discarded. The pendulum formula depends on the first of them. For small swings the restoring force involves $\sin \theta$, and replacing $\sin \theta$ by θ is exactly what turns the equation of motion into simple harmonic motion, which can then be solved.

The series also prices the approximation. The first term discarded is $\frac{\theta^3}{6}$, so that is roughly the error, and the relative error stays under 1% for swings up to about 14° . Beyond that the pendulum's real period drifts away from the textbook formula, by an amount large enough to measure.

A Maclaurin series can also be built directly *from a differential equation*, by differentiating the equation itself to obtain the derivatives at 0. This joins the two halves of the chapter.

EXAMPLE 16.12

A function satisfies $\frac{dy}{dx} = 1 + y^2$ with $y(0) = 0$. Find the Maclaurin series of y up to the term in x^3 .

Take the derivatives at 0 from the equation itself. First, $y'(0) = 1 + 0 = 1$. Differentiating the equation,

$$y'' = 2y y' \implies y''(0) = 0, \quad y''' = 2(y')^2 + 2y y'' \implies y'''(0) = 2.$$

Feed these into the Maclaurin formula:

$$y = 0 + x + \frac{0}{2!}x^2 + \frac{2}{3!}x^3 + \cdots = x + \frac{x^3}{3} + \cdots$$

You may recognise the answer: $y = \tan x$ solves this equation, since $\frac{d}{dx} \tan x = \sec^2 x = 1 + \tan^2 x$, and these are indeed the first terms of its series. The equation was never solved. Each derivative at 0 was read off from the equation itself, one at a time, and that was enough to build the series.

TIP Maclaurin series suggest several good IA topics. Two worth exploring: Madhava's series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$, which comes from setting $x = 1$ in the arctangent series, and why it converges so slowly that a better choice such as $\arctan \frac{1}{\sqrt{3}}$ reaches the same accuracy in a fraction of the terms; or how a calculator really evaluates $\sin x$, by reducing the angle to a small interval first and then using a short truncated series.

EXAMPLE 16.13

Use Maclaurin series to evaluate $\lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2}$.

Expand the numerator with the series for e^x :

$$e^x - 1 - x = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots\right) - 1 - x = \frac{x^2}{2} + \frac{x^3}{6} + \cdots$$

Divide by x^2 :

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + \frac{x}{6} + \cdots \xrightarrow{x \rightarrow 0} \frac{1}{2}.$$

The series settles an indeterminate limit as neatly as L'Hôpital's rule, and it shows *why* the answer is $\frac{1}{2}$: that number is the leading coefficient.

YOUR TURN 16.4 Maclaurin Series

7. Find the first three nonzero terms of the Maclaurin series for each function.

(a) e^{2x}

(b) $\cos x$

(c) $\ln(1 + 2x)$

(d) e^{-x^2}

8. Find the first two nonzero terms of the Maclaurin series for each function.

(a) $\sin 2x$

(b) $\sin(x^2)$

(c) $\arctan 2x$

(d) $\sqrt{1+x}$

9.

- (a) Use the first three terms of the series for e^x to estimate $e^{0.1}$.
- (b) Use the first three terms of the cosine series to estimate $\cos 0.2$ to four decimal places.
- (c) Compare each estimate with the true value, and state which is the more accurate. Explain why.
- (d) Find the first three nonzero terms of the Maclaurin series for $e^x \cos x$.
- (e) Differentiate the series $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ term by term, and identify the series you obtain.
- (f) Differentiate the series for $\ln(1+x)$ term by term, and verify that the result is the expansion of $\frac{1}{1+x}$.

10. A function satisfies $\frac{dy}{dx} = x + y$ with $y(0) = 1$.

- (a) Write down $y'(0)$.
- (b) By differentiating the equation, find $y''(0)$ and $y'''(0)$.
- (c) Hence find the Maclaurin series for y up to the term in x^3 .
- (d) Use the series to estimate $y(0.3)$, and compare it with the Euler estimate 1.362 from the notes.

16.5 Euler's Method

Not every differential equation can be solved exactly, and most of the equations that describe the real world cannot. When no exact solution exists, the solution can still be approximated numerically. **Euler's method** does this by walking along the curve in short straight steps, taking the gradient of each step from the differential equation itself.

The idea is the tangent line of Ch. 13. Near a known point the curve is close to its tangent, so moving a small distance h along the tangent lands close to the curve:

$$y \approx y_0 + \underbrace{f(x_0, y_0)}_{\text{gradient}} \times \underbrace{h}_{\text{step}}.$$

The differential equation supplies the gradient at whatever point you have reached, so the same move can be repeated from the new point, and again from the one after that.

KEY · IB

$$y_{n+1} = y_n + h f(x_n, y_n), \quad x_{n+1} = x_n + h$$

Here h is the fixed **step length**. Each step is exact only at its starting point and drifts from

the curve thereafter, so shorter steps stay closer, at the cost of needing more of them.

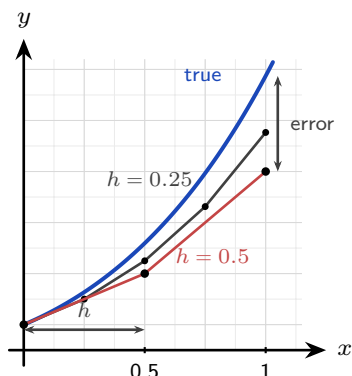


Figure 16.2 Euler's method on $\frac{dy}{dx} = x + y$ with $y(0) = 1$. Each straight step leaves the curve, which bends upward away from it, so the estimate falls short and the gap at $x = 1$ is the error. Halving the step length roughly halves that gap.

The figure shows two things at once. Each step is a straight line, so it leaves a curve that bends away from it, and the gap at the far end is the *error*. The gray path takes steps half as long and stays about twice as close to the curve. Shorter steps give more accuracy, at the cost of more of them.

EXAMPLE 16.14

Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's method with $h = 0.1$ to estimate $y(0.3)$.

Apply $y_{n+1} = y_n + h(x_n + y_n)$ three times. A table is the clearest way to organise the work, and it is what an examiner expects to see.

n	x_n	y_n	$f = x_n + y_n$	$h f$
0	0	1	1	0.1
1	0.1	1.1	1.2	0.12
2	0.2	1.22	1.42	0.142
3	0.3	1.362		

Each y_n is the previous one plus the $h f$ on the line above. So $y(0.3) \approx 1.362$, against an exact value of about 1.3997.

Euler has undershot, and the reason is visible in the figure: this curve is concave up, so every tangent lies below it and each step falls short. That is the general rule. A concave-up solution is underestimated and a concave-down one is over-estimated, so the sign of $\frac{d^2y}{dx^2}$ tells you which way your answer is wrong before you check it.

The method matters most when no exact solution is available. Make the resistance on a falling body depend on the shape of the object and on the air density, which changes with height, and the equation has no solution among the functions of this course, but Euler's method still produces a table of values. This is, in miniature, how weather forecasts are com-

puted: equations too hard to solve exactly, stepped forward numerically, millions of times.

EXAMPLE 16.15

A skydiver falls from rest, and air resistance gives $\frac{dv}{dt} = 9.8 - 0.1v^2$ (in m s^{-1} and seconds). Use Euler's method with $h = 0.5$ to estimate the speed after 1.5 seconds, and compare with the terminal velocity.

Apply $v_{n+1} = v_n + 0.5(9.8 - 0.1v_n^2)$ three times from $v_0 = 0$:

$$v_1 = 0 + 0.5(9.8) = 4.9$$

$$v_2 = 4.9 + 0.5(9.8 - 0.1(4.9)^2) = 4.9 + 0.5(7.399) = 8.60$$

$$v_3 = 8.60 + 0.5(9.8 - 0.1(8.60)^2) = 8.60 + 0.5(2.404) = 9.80$$

So $v(1.5) \approx 9.80 \text{ m s}^{-1}$. The terminal velocity is where acceleration dies: $9.8 - 0.1v^2 = 0 \implies v = \sqrt{98} \approx 9.90 \text{ m s}^{-1}$. After only three steps the estimate is already within 1% of the terminal speed. The physics is visible in the arithmetic: each bracket $9.8 - 0.1v_n^2$ is the net force per unit mass, and it shrinks towards zero as the speed builds.

YOUR TURN 16.5 Euler's Method

11. Use Euler's method for each of the following.

- (a) $\frac{dy}{dx} = 2x$, $y(0) = 1$; use $h = 0.5$ to estimate $y(1)$.
- (b) $\frac{dy}{dx} = y$, $y(0) = 1$; use $h = 0.1$ to estimate $y(0.2)$.
- (c) $\frac{dy}{dx} = x + y$, $y(0) = 1$; use $h = 0.2$ to estimate $y(0.4)$.
- (d) $\frac{dy}{dx} = xy$, $y(1) = 1$; use $h = 0.1$ to estimate $y(1.2)$.

12.

- (a) For $\frac{dy}{dx} = xy$ with $y(1) = 2$, take one step with $h = 0.2$ to estimate $y(1.2)$.
- (b) For $\frac{dy}{dx} = x + y$ with $y(0) = 0$, take two steps with $h = 0.5$ to estimate $y(1)$.
- (c) Solve $\frac{dy}{dx} = y$ with $y(0) = 1$ exactly, and compare the exact $y(0.2)$ with your answer to question 11 part (b).
- (d) State whether Euler's method over- or under-estimates the solution of part (c), and explain why in terms of the shape of the curve.

Reflections

TOK Euler's method rests on a striking idea: knowing the rule for change at every point, plus a single starting state, determines the entire future of a system. This is mathematical determinism. Yet the same equations can be so sensitive to their starting values that long-term prediction becomes impossible in practice, the phenomenon of chaos. Determined but unpredictable: what does that tell us about the limits of knowledge?

TOK A truncated series is deliberately wrong. It is chosen because it is simple enough to compute, and every extra term bought accuracy at the cost of that simplicity: four terms of the sine series give $\sin 1$ correct to five decimal places, and one term gives it to none.

Is there always a trade-off between accuracy and simplicity? And when a model is knowingly approximate, as every model in this chapter is, what makes it good enough to act on?

IM Both halves of this chapter have a long international lineage.

The series came first from India. Mathematicians of the Kerala school, notably Madhava in the 14th century, found the expansions of $\sin x$, $\cos x$ and $\arctan x$ roughly three centuries before Newton, Leibniz and Maclaurin. The arctangent series, usually named after Gregory and Leibniz, was known in Kerala generations earlier.

Numerical methods have a similar story in the Islamic world. In the 15th century Jamshīd al-Kāshī, working at the observatory in Samarkand, needed $\sin 1^\circ$ to extraordinary precision for his astronomical tables. No formula gave it, so he wrote down an equation the value satisfied and solved it by repeated approximation, each round feeding its answer into the next, reaching sixteen correct decimal places. That is the same bargain Euler's method strikes: when no exact answer is available, iterate towards one and control the error. Two centuries earlier Sharaf al-Dīn al-Ṭūsī had studied when a cubic equation has solutions by examining where its maximum lies, an argument that reads today like the use of a derivative.

Mathematical discovery is rarely the work of one place or one name.

RW Differential equations are the language in which physics is written. Newton's second law $F = ma$ is a differential equation for position; radioactive decay, population growth, electrical circuits, epidemics, and planetary orbits are all governed by them. To predict a system's future is, almost always, to solve a differential equation, exactly when you can and numerically (by methods like Euler's) when you cannot.

EXT The five standard series behave differently at the edges. The series for e^x , $\sin x$ and $\cos x$ converge for every x , but $\arctan x$ only for $|x| \leq 1$ and $\ln(1+x)$ only for $-1 < x \leq 1$, while the geometric series they are built from needs $|x| < 1$. Each series has a *radius of convergence*, and outside it the polynomial does not merely lose accuracy: it diverges completely. Finding that radius, and explaining why $\ln(1+x)$ has one at all when e^x does not, is the beginning of complex analysis, where the answer turns out to depend on where the function misbehaves in the complex plane. See Ch. 17.

EXT Maclaurin series turn calculus into algebra: integrals with no elementary antiderivative, like $\int e^{-x^2} dx$, can be integrated term by term once the function is a series. Series also solve differential equations that no other method in this course can reach, by assuming the answer is a power series and matching coefficients. The infinite polynomial is one of the most far-reaching ideas in all of analysis.

End-of-Chapter Problems

1. A population grows so that $\frac{dP}{dt} = 0.2P$, with $P = 500$ when $t = 0$.
 (a) Solve the differential equation for $P(t)$. [4]
 (b) Find the population when $t = 10$. [2]
2. A body cools according to $\frac{d\theta}{dt} = -0.1(\theta - 25)$, with $\theta(0) = 90$ (in $^{\circ}\text{C}$).
 (a) Solve for $\theta(t)$. [4]
 (b) Find the temperature after 10 minutes. [2]
 (c) State the temperature the body approaches in the long run. [1]
3. Consider $\frac{dy}{dx} = \frac{x^2 + y^2}{xy}$ for $x > 0$.
 (a) Use the substitution $y = vx$ to show that $xv \frac{dv}{dx} = 1$. [4]
 (b) Hence find the general solution. [3]
4. Consider $\frac{dy}{dx} + \frac{1}{x}y = 3x$ for $x > 0$.
 (a) Find the integrating factor. [2]
 (b) Solve the equation given $y(1) = 2$. [4]
5. Consider $\frac{dy}{dx} = x + y$, with $y(0) = 1$.
 (a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$, showing each step. [4]
 (b) Solve the equation exactly (integrating factor) and find the true $y(0.5)$. [4]
 (c) Comment on the size and direction of the error. [2]
6. Let $f(x) = \cos x$.
 (a) Find $f(0), f'(0), f''(0), f'''(0), f^{(4)}(0)$. [3]
 (b) Hence write the Maclaurin series of $\cos x$ up to the term in x^4 . [2]
 (c) Use it to estimate $\cos(0.2)$. [2]
7. EXACT
 (a) Write down the Maclaurin series for $\sin x$ up to the term in x^5 . [2]
 (b) Hence evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$. [3]

8.

- (a) Write down the Maclaurin series for e^x up to the term in x^3 . [1]
 (b) By substituting $-x^2$, find the series for e^{-x^2} up to x^4 . [2]
 (c) Hence estimate $\int_0^1 e^{-x^2} dx$ using terms up to x^4 . [4]

9. A bowl of miso soup at 90°C is left in a 25°C room and obeys $\frac{d\theta}{dt} = -k(\theta - 25)$. After 10 minutes it has cooled to 65°C .

- (a) Solve the differential equation, expressing θ in terms of k and t . [4]
 (b) Find k . [2]
 (c) Find how long, from the start, the soup takes to reach 40°C . [2]

10. Consider $\frac{dy}{dx} = \frac{y^2 + xy}{x^2}$ for $x > 0$, with $y(1) = 1$.

- (a) Use the substitution $y = vx$ to show that $x \frac{dv}{dx} = v^2$. [3]
 (b) Hence solve for y in terms of x . [4]

11. Solve each linear equation.

- (a) $\frac{dy}{dx} - \frac{y}{x} = x^2$ for $x > 0$, given $y(1) = 1$. [6]
 (b) $\frac{dy}{dx} + 2y = e^{-x}$ with $y(0) = 0$. [6]

12. Solve $\frac{dy}{dx} + (\tan x)y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$, given $y(0) = 2$. [7]

13. A fish population n (in thousands) in a lake satisfies the logistic equation $\frac{dn}{dt} = 0.4n(2 - n)$, with $n(0) = 0.5$.

- (a) Using partial fractions, show that the solution is $n = \frac{2}{1 + 3e^{-0.8t}}$. [6]
 (b) Find the time at which the population reaches 1500 fish. [3]

14. Consider $\frac{dy}{dx} = y - x^2$ with $y(0) = 1$.

- (a) Use Euler's method with $h = 0.25$ to estimate $y(0.5)$. [5]
 (b) The exact solution is concave up on $0 \leq x \leq 0.5$. State, with a reason, whether your estimate is above or below the true value. [2]

15. A skydiver falls from rest with $\frac{dv}{dt} = 9.8 - 0.2v^2$.

- (a) Use Euler's method with $h = 0.5$ to estimate $v(1.5)$. [4]

- (b) Find the terminal velocity, and comment on the behaviour of the Euler estimates around it. [3]

16. Find the Maclaurin series of $\ln(1 + 2x)$ up to the term in x^3

- (a) by substitution into a standard series; [3]
 (b) by computing derivatives from the definition. [3]

17.

- (a) Find the Maclaurin series of $e^{-x} \sin x$ up to the term in x^3 . [6]
 (b) Use your series to estimate $e^{-0.1} \sin 0.1$, and compare it with the value from your calculator. [3]

18. By setting $x = 1$ in the Maclaurin series for $\arctan x$:

- (a) write down a series for $\frac{\pi}{4}$; [2]
 (b) estimate π using the first four terms; [2]
 (c) comment on the rate of convergence. [1]

19. EXACT Use Maclaurin series to evaluate $\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3}$. [5]

20. A function satisfies $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$. Find the Maclaurin series of y up to the term in x^3 . [7]

21. A drug is infused into the bloodstream at a rate that raises its concentration by r mg L^{-1} per hour, and the body removes it at a rate proportional to the concentration C , so that

$$\frac{dC}{dt} + kC = r,$$

where t is measured in hours and $C(0) = 0$.

- (a) Explain why this equation is linear, and write down its integrating factor. [3]
 (b) Solve the equation to show that $C = \frac{r}{k} (1 - e^{-kt})$. [4]
 (c) A patient is given $r = 6$ and has $k = 0.3$. Find the concentration after 4 hours. [2]
 (d) Find the concentration the patient approaches in the long run, and explain what this means for the treatment. [3]
 (e) Find how long it takes to reach 90% of that level. [2]
 (f) Once that level is reached the infusion is stopped, so that $\frac{dC}{dt} = -kC$. Solve this equation and find how long the concentration then takes to halve. [3]

22. In this question you will build a series for logarithms that converges quickly.

- (a) Find the Maclaurin series of $\ln(1+x)$ up to the term in x^4 . [4]
 (b) Write down the Maclaurin series of $\ln(1-x)$ up to the term in x^4 . [2]
 (c) Hence show that $\ln\left(\frac{1+x}{1-x}\right) = 2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \dots\right)$. [3]
 (d) By choosing a suitable value of x , use the three terms shown to estimate $\ln 3$. [3]
 (e) The series for $\ln(1+x)$ is valid only for $-1 < x \leq 1$. State the value of x that would be needed to reach $\ln 3$ from that series alone, and explain why the series in part (c) is the more useful one here. [2]

23. Water drains from a tank through a hole in its base. The depth h metres at time t minutes satisfies

$$\frac{dh}{dt} = -k\sqrt{h},$$

where k is a positive constant.

- (a) Separate the variables to show that $2\sqrt{h} = c - kt$ for some constant c . [3]
 (b) Given that the depth is h_0 when $t = 0$, show that $h = \left(\sqrt{h_0} - \frac{1}{2}kt\right)^2$. [2]
 (c) A tank starts with $h_0 = 1.44$ and has $k = 0.2$. Find how long it takes to empty. [3]
 (d) Find the depth after 6 minutes. [2]
 (e) Use Euler's method with steps of 3 minutes to estimate the depth after 6 minutes, and find the error in this estimate. [4]
 (f) Describe the shape of the graph of h against t , and use it to explain why the Euler estimate lies below the exact value. [2]

Challenge Questions

24. EXACT **How fast does coffee cool?** The chapter opened with $\frac{d\theta}{dt} = -k(\theta - 20)$, where θ is the temperature in degrees and t the time in minutes. A cup is poured at 80° , so $\theta(0) = 80$.

- (a) Solve the equation to show that $\theta = 20 + 60e^{-kt}$. [3]
 (b) The coffee has cooled to 60° after 10 minutes. Find k exactly, and to three significant figures. [3]
 (c) Find the temperature after 20 minutes. [2]
 (d) Show that the time taken to reach a temperature θ is $t = \frac{1}{k} \ln\left(\frac{60}{\theta - 20}\right)$, and explain why the coffee never reaches 20° . [4]
 (e) A second cup is poured at 90° in the same room. Show that it takes exactly 10 minutes to cool from 80° to 60° , the same as the first cup. [3]

25. **How wrong is Euler?** Take $\frac{dy}{dx} = y$ with $y(0) = 1$, whose exact solution is $y = e^x$, so

that $y(1) = e$.

- (a) Show that one Euler step of length h gives $y_{n+1} = (1 + h)y_n$, and hence that after n steps $y_n = (1 + h)^n$. [3]
- (b) Taking $h = \frac{1}{n}$, deduce that Euler's estimate of $y(1)$ is $\left(1 + \frac{1}{n}\right)^n$. [2]
- (c) Evaluate this estimate for $n = 2, 4, 8, 16$, and tabulate the error in each case. [4]
- (d) Describe what happens to the error each time h is halved, and state what this says about the accuracy of the method. [3]
- (e) Write down the limit of $\left(1 + \frac{1}{n}\right)^n$ as $n \rightarrow \infty$, and explain why Euler's method had to give this answer. [3]

26. The ferry and the current. A ferry sets off from $(1, 0)$ toward a dock at the origin, always aiming straight at the dock, while a river current pushes it in the positive y -direction at a fraction k of the ferry's own speed. Its path satisfies

$$\frac{dy}{dx} = \frac{y - k\sqrt{x^2 + y^2}}{x}.$$

- (a) Show the equation is homogeneous, and that $y = vx$ leads to $x \frac{dv}{dx} = -k\sqrt{1 + v^2}$. [3]
- (b) Given that $\int \frac{dv}{\sqrt{1 + v^2}} = \ln(v + \sqrt{1 + v^2}) + C$, show that $v + \sqrt{1 + v^2} = x^{-k}$. [3]
- (c) Hence show that $y = \frac{1}{2}(x^{1-k} - x^{1+k})$. [4]
- (d) Taking $k = \frac{1}{2}$, find where the ferry lands. [2]
- (e) Show that the ferry reaches the dock only when $k < 1$, and describe what happens when $k = 1$ and when $k > 1$. [4]

